

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: CSA51 **COURSE NAME:** Cryptography and Network Security

COURSE OUTCOME COVERED

CO4: *Implement best security practices for IP, email, and web service using PGP, S/MIME for enhancing email Security.CO (BL3, BL6)*

Assessment Tool Weightage: 10%

QUESTIONS: 5

TOTAL MARKS: 25

Annexure A

Comparative Analysis

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Comparative Analysis

1. Compare Authentication Header (AH) and Encapsulating Security Payload (ESP) in IPSec with respect to security services, performance overhead, and suitability for different network scenarios. Analyze which is more appropriate for a VPN deployment.
2. Compare transport mode and tunnel mode in IPSec architecture. Evaluate their effectiveness in securing host-to-host versus network-to-network communication.
3. Perform a comparative analysis of Pretty Good Privacy (PGP) and S/MIME in terms of key management, trust models, scalability, and enterprise usability. Recommend the most suitable approach for a corporate email system.
4. Compare rule-based and machine learning-based email spam detection techniques. Analyze their effectiveness in handling evolving spam patterns and minimizing false positives.
5. Compare manual key management and automated key management (e.g., using IKE) in IPSec. Evaluate their impact on security, scalability, and operational complexity in large organizations.

RUBRICS FOR CALCULATION:

Q. No	Understanding of Concepts (25)	Experimental Design (30)	Use of Tools (20)	Analysis of Results (15)	Documentation (10)	Total Score (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	
5	/ 4	/ 4	/ 4	/ 4	/ 4	
OVERALL RUBRICS VALUE						
	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 25	/ 30	/ 20	/ 15	/ 10	/ 100

Cryptography CO4AT3:

R. Deepasri (192511309)

Comparative Analysis of IPSec, Email Security, and Spam Detection

1. AH vs ESP in IPSec

Feature	AH (Authentication Header)	ESP (Encapsulating Security Payload)
Main security service	Authentication, integrity, and anti-replay	Confidentiality, integrity, authentication, and anti-replay
Encryption	✗ No	✓ Yes
Data integrity	✓ Yes	✓ Yes
Data authentication	✓ Yes	✓ Yes
Confidentiality	✗ No	✓ Yes
Overhead	Lower than ESP when encryption is not required	Higher because of encryption and additional processing
NAT compatibility	Poor; AH authenticates parts of the IP header that NAT may modify	Better, especially with NAT-T
Typical use	Situations requiring authentication/integrity without encryption	VPNs, confidential communications, secure Internet traffic

Analysis:

AH is useful when data integrity and authentication are required but confidentiality is unnecessary. However, ESP is generally preferred because it provides both confidentiality and integrity. For a **VPN deployment, ESP is more appropriate**, particularly ESP with authenticated encryption, because VPN traffic normally needs to be protected from both modification and eavesdropping.

2. Transport Mode vs Tunnel Mode

Feature	Transport Mode	Tunnel Mode
Protection	Protects the IP payload	Protects the entire original IP packet
New IP header	No	Yes
Overhead	Lower	Higher
Performance	Faster	Slightly slower
Host-to-host	Highly suitable	Can also be used
Network-to-network	Less suitable	Highly suitable
VPN gateways	Less common	Common
Privacy of internal IP addresses	Limited	Better

Host-to-host communication:

Transport mode is suitable when two individual hosts communicate directly. It provides security while keeping the original IP header visible.

Network-to-network communication:

Tunnel mode is better for site-to-site VPNs because the complete original IP packet is encapsulated and protected. A new IP header is added for routing between VPN gateways.

Conclusion:

- Host-to-host → Transport mode
- Network-to-network/VPN → Tunnel mode

3. PGP vs S/MIME

Feature	PGP	S/MIME
Key management	Users commonly manage their own keys	Usually managed through certificates and CAs
Trust model	Web of Trust	Hierarchical Certificate Authority (CA)

Feature	PGP	S/MIME
Scalability	More difficult in large organizations	Highly scalable
Enterprise integration	Moderate	Strong
Digital signatures	✓ Yes	✓ Yes
Encryption	✓ Yes	✓ Yes
Certificate management	Less centralized	Centralized
User control	High	More organization-controlled
Corporate usability	Moderate	High

Analysis:

PGP provides a decentralized **Web of Trust**, where users can verify and sign each other's keys. This provides flexibility but can become difficult to manage as the organization grows.

S/MIME uses digital certificates issued by trusted Certificate Authorities. Organizations can centrally manage certificates, policies, and user identities. This makes S/MIME more suitable for large enterprises.

Recommendation:

For a **corporate email system**, **S/MIME is generally more suitable** because of centralized key management, scalability, certificate-based authentication, and integration with enterprise email environments.

4. Rule-Based vs Machine Learning-Based Spam Detection

Feature	Rule-Based Detection	Machine Learning-Based Detection
Detection method	Predefined rules	Learns patterns from data
Implementation	Simple	More complex
Adaptability	Low	High

Feature	Rule-Based Detection	Machine Learning-Based Detection
New spam patterns	Difficult to detect	Better at detecting evolving patterns
False positives	Can be high if rules are too strict	Can be reduced with good training
Maintenance	Rules require manual updates	Model can be retrained
Training data	Not required	Required
Computational requirement	Low	Higher
Scalability	Limited for complex patterns	Better for large-scale systems

Analysis:

Rule-based systems use conditions such as suspicious keywords, sender information, URLs, or message patterns. They are simple and fast but struggle when attackers change their techniques.

Machine learning systems learn characteristics of legitimate and spam messages. Algorithms such as Naive Bayes, decision trees, and neural networks can identify complex patterns and adapt to changing spam behavior.

Conclusion:

For modern email systems, **machine learning is generally more effective** because spam patterns continuously evolve. However, combining **rule-based filtering + machine learning** can provide stronger protection and help reduce false positives.

5. Manual Key Management vs Automated Key Management

Feature	Manual Key Management	Automated Key Management (IKE)
Key generation	Manually configured	Automatically negotiated
Scalability	Poor	Excellent

Feature	Manual Key Management	Automated Key Management (IKE)
Security	More vulnerable to configuration errors	Generally stronger
Operational complexity	Very high for large networks	Lower
Key rotation	Manual	Automatic
Authentication	Manually configured	Negotiated through IKE
Large organizations	Difficult	Highly suitable
Administration	Time-consuming	More centralized/automated
Risk of human error	High	Lower

Analysis:

Manual key management may be practical for a small network with only a few IPSec connections. However, as the number of devices increases, manually creating, distributing, and changing keys becomes difficult and error-prone.

IKE (Internet Key Exchange) automates security association negotiation, authentication, and key establishment. It also supports automatic key renewal, making it much more practical for large organizations.

Conclusion:

For a large enterprise, **automated key management using IKE is strongly preferred** because it improves scalability, reduces administrative effort, supports key rotation, and minimizes configuration errors.

Overall Comparative Conclusion

Area	Better Choice	Reason
IPSec security	ESP	Provides confidentiality plus integrity/authentication
Host-to-host IPSec	Transport Mode	Lower overhead and protects the payload
Network-to-network VPN	Tunnel Mode	Protects the complete original IP packet

Area	Better Choice	Reason
Corporate email security	S/MIME	Centralized certificate and key management
Modern spam detection	Machine Learning	Adapts better to changing spam patterns
Large-scale IPSec key management	IKE	Automated, scalable, and reduces human error

Final conclusion: For a modern enterprise environment, **ESP in tunnel mode with IKE-based automated key management** is the preferred IPSec solution for VPNs. For corporate email, **S/MIME** provides strong centralized management, while **machine-learning-based spam detection**, ideally combined with rules, offers better protection against evolving spam.